Matteo Torres

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Website | GitHub | LinkedIn

EDUCATION

Master of Environmental Data Science (MEDS) (06/25)
Bren School of Environmental Science & Management — University of California, Santa Barbara (UCSB)

Highlighted Coursework: Databases and Data Management, Geospatial Analysis and Remote Sensing, Data Visualization and Communication, Python for Environmental Data Science
Awards: Environmental Data Science Fellowship

Bachelor of Science in Biology | Minor in Dance (12/23)
College of Natural Sciences & Mathematics — California State University, Long Beach (CSULB)

Highlighted Coursework: Geographic Information Science for Natural Sciences, Conservation Biology, Population Ecology, Biostatistics, Undergraduate Directed Research
Awards: Gulf of Maine Research Institute Research Grant, NSF Primate Population Ecology Research Grant

DATA SCIENCE PROJECTS

Assessing Range Shifts of Coastal Species to Inform Conservation in a Key California Biogeographic Transition Zone
Master's Capstone Project
Repository | Technical Documentation | Dashboard

- Mapped the habitat range distributions of rocky intertidal species along the California coastline in 100 *km* coastal segments to identify species or species groups with range edges at or near Point Conception
- Utilized generalized additive models (GAMs) to analyze contemporary range shifts of northern and southern boundaries at five-year intervals from 2000 to 2024, identifying range extensions or contractions over time
- Projected habitat suitability for the year 2050 using an ensemble species distribution model (ESDM) approach under varying environmental conditions and a SSP2-4.5 climate scenario, highlighting areas of habitat gain or loss
- Developed an interactive Shiny web application called *California Ranges of Intertidal Species Portal (CRISP)*, which compiles all deliverables into a priority monitoring assessment for conservation decisions

Pokémon Database
Database Management Project
Repository | Data Visualizations

- Gathered and cleaned multiple Kanto Pokémon datasets in R to ensure seamless integration into a relational database
- Designed and built a relational Pokémon database using SQL and DuckDB, integrating datasets to capture detailed attributes for each Pokémon, similar to a Pokédex
- Queried the Pokémon database using R and Python to extract insights and create engaging data visualizations

Additional Projects: Thomas Fire Analysis (Repository, Blog), Spotify Wrapped (Shiny, Tableau), Exploring Historical Redlining (Repository), Biodiversity Intactness Index (BII) Change (Repository), Identifying Impacts of Extreme Weather (Repository)

RESEARCH EXPERIENCE

Primate Population Ecology Research Internship — CSULB Quantitative Ecology Lab (05/23 — 12/23)
Maternal Inexperience and Social Attention in Rhesus Macaques

- Merged and cleaned annual census and socioemotional response datasets, grouping rhesus macaques by life stage: juvenile (0-5 *yrs.*), adult (6-14 *yrs.*), senescent (15+ *yrs.*)
- Developed linear mixed-effects models to analyze looking time, incorporating sex, life stage, and experimental variables as fixed effects, and individual and maternal ID as random effects
- Ran ANOVAs to test the significance of fixed effects and applied likelihood ratio tests to compare the base model with one including the adversity variable

Inclusive NOAA Fisheries (INFISH) Internship Program — University of Maryland Eastern Shore (06/21 — 08/21)
Connections Between Low-Level Arctic Clouds and Extreme Winter Precipitation

- Gathered and organized LiDAR NRB cloud signal profiles for Fairbanks, Alaska, up to 5 *km* altitude, during the winter season (October – March) from 2019 to 2021
- Processed NRB profiles by calculating the first derivative of the cloud signal to identify its boundaries, cloud base height and cloud top, and derive cloud thickness
- Isolated extreme precipitation events, where changes in precipitation occurrence and intensity were detected, and compared cloud signal boundaries and thickness statistics between normal and extreme periods

SKILLS

Technical: R, Python, SQL, Bash, ArcGIS Pro, Git/GitHub, Shiny, Tableau, Canva, Microsoft Office Suite, Google Workspace